

Python - for Loops & range()

Comprehensive Lesson Plan & Curriculum Guide (Grade 3)

Subject: Computer Science / Python Programming	Target Grade: Grade 3 (Ages 8–9)
Core Concepts: Repetition, For Loops, range () function, List Iteration	Duration: 60 Minutes

1. Lesson Overview & Objectives

This lesson introduces primary school students to the foundational programming concept of **Loops** using Python. Instead of manually writing repetitive commands, students will discover how a computer automates sequential actions efficiently using for loops and the range () function.

Learning Objectives

- **Understand Repetition:** Explain what a loop is and identify why repetitive manual code is inefficient.
- **Syntax Mastery:** Write and execute basic Python for loops using valid syntax (including colons and indentations).
- **The range() Function:** Master utilizing range () with one parameter (stop), two parameters (start, stop), and three parameters (start, stop, step).
- **List & Backwards Iteration:** Loop through collections (lists) and implement reverse countdown sequences.

2. Introduction: What is a Loop?

A **loop** repeats a block of code multiple times. In general programming, instead of writing the exact same line of code over and over again, we use a loop to keep our instructions clean and concise.

The Power of Loops

Imagine your teacher asks you to write the word 'Hello' 100 times on a sheet of paper! It would take a long time and your hand would get very tired. In Python, a loop can complete this massive task using just **2 lines of code!**

3. The Python for Loop Syntax

The `for` loop is used to iterate over a sequence of values. Let's examine a core example:

```
for i in range(5):  
    print(i)
```

Output: 0, 1, 2, 3, 4

How a for Loop Executes Step-by-Step

1. **Pick Value:** Python selects the first value available in the designated sequence.
2. **Run Code:** It executes the indented block of code directly underneath the loop declaration.
3. **Go Back:** It loops back up to select the next sequence value.
4. **Repeat:** It continues this automatic loop until all values in the sequence have been processed!

4. Deep Dive into the `range()` Function

The `range()` function generates an orderly sequence of numbers. It acts as an automated guide, telling the loop exactly how many times it needs to repeat.

Syntax Example	Generated Sequence	Behavior Rules
<code>range(5)</code> (One Number)	0, 1, 2, 3, 4	Starts automatically at 0 and stops right before the number 5 (5 total numbers).
<code>range(1, 6)</code> (Two Numbers)	1, 2, 3, 4, 5	First number = Start (included). Second number = Stop (not included).
<code>range(2, 7)</code> (Two Numbers)	2, 3, 4, 5, 6	Starts counting at 2 and stops directly before 7.
<code>range(0, 10, 2)</code> (Three Numbers)	0, 2, 4, 6, 8	The third number represents the Step . It skips numbers along the way!

Counting Backwards (The Countdown Loop)

We can also count downwards by configuring a negative step argument. This is perfect for simulating a rocket ship countdown!

```
for i in range(5, 0, -1):  
    print(i)
```

Output: 5, 4, 3, 2, 1
(Countdown!)

5. Iterating Through Lists

Loops are incredibly useful for handling collections of items, such as lists of data. For example, if we have a basket filled with tasty fruits, we can easily loop through each individual entry:

```
fruits = ['mango', 'banana', 'apple']  
for f in fruits:  
    print(f)
```

Output:
mango
banana
apple

6. Fun with Code Patterns: Building Stars

By combining strings with loop index values, we can draw structural geometric shapes like text-based triangles on screen:

```
for i in range(1, 6):  
    print('*' * i)
```

Output:
*
* *
* * *
* * * *
* * * * *

7. Common Mistakes to Avoid

When starting out with Python loops, keep an eye out for these frequent slip-ups:

Watch Out For These Loop Bugs!

- **Forgetting the Colon (:):** Every loop definition line must end with a colon. Without it, Python will flag a syntax error!
- **Not Indenting the Loop Body:** All instructions belonging inside the loop must be pushed to the right using the Tab key or spaces.
- **Wrong Range Bounds:** Remember that the stop number is *never included* in the loop's output stream.

8. Classroom Challenges & Activities

Put your new coding skills to the test with these practical exercises:

Challenge 1: The Name Repeater

Write a script that prints your own name 5 times sequentially on screen.

Challenge 2: Even Number Explorer

Create a loop using a step argument to print all even numbers starting from 2 up to and including 20.

Challenge 3: The Hashtag Triangle

Modify the star triangle structure code to display an escalating pyramid using the hashtag character (#).

Great Job! Keep Coding!

You have officially mastered Python for loops and the `range()` utility! Now you can make Python repeat anything you can think of. Keep practicing and have fun building scripts!